FOUNDATIONAL IT SKILLS

Student Notes

Experiments 3 & 4

Web & Connectivity

Professional Documentation

For Polytechnic Diploma Students

Computer Laboratory

Contents

1 Experiment 3 – Web & Connectivity **3**
1.1 Learning Objectives . . . 3
1.2 What is the Internet? . . . 3
1.3 What is the World Wide Web? . . . 4
1.4 What is a Web Browser? . . . 4
1.5 What is a Search Engine? . . . 5
1.6 Web Browser vs Search Engine . . . 5
1.7 What is Advanced Search? . . . 5
1.8 The `filetype:` Search Operator . . . 6
1.9 The `site:` Search Operator . . . 6
1.10 Using Multiple Search Operators . . . 7
1.11 Use of Quotation Marks in Search . . . 7
1.12 Practical Activity – Advanced Search . . . 8
1.13 What is Email? . . . 8
1.14 What is Email Etiquette? . . . 9
1.15 Structure of a Professional Email . . . 9
1.16 Recipient . . . 9
1.17 Subject . . . 10
1.18 Greeting . . . 10
1.19 Body . . . 10
1.20 Closing . . . 10
1.21 Example of a Professional Email . . . 11
1.22 Common Email Mistakes . . . 11
1.23 Email Activity . . . 12
1.24 What is an LMS? . . . 12
1.25 Google Classroom . . . 13
1.26 Summary – Web & Connectivity . . . 13

2 Experiment 4 – Professional Documentation **15**
2.1 Learning Objectives . . . 15
2.2 What is a Document? . . . 15
2.3 What is Professional Documentation? . . . 15
2.4 Formal Report . . . 16
2.5 What are Styles? . . . 16
2.6 Why are Styles Important? . . . 17
2.7 Heading Styles . . . 17
2.8 Manual Formatting vs Heading Style . . . 18

2.9 What is a Table of Contents? 18
2.10 Why is a Table of Contents Useful? 19
2.11 Relationship Between Styles and Table of Contents 19
2.12 Creating a Simple Formal Report 20
2.13 Practical Activity – Professional Documentation 21
2.14 Practical Procedure 21
2.15 Copy and Move 22
2.16 Common Mistakes in Professional Documentation 23
2.17 Combined Practical Activity 23

3 Quick Revision Questions **25**
3.1 Web & Connectivity 25
3.2 Professional Documentation 25

4 Quick Summary **26**
4.1 Experiment 3 – Web & Connectivity 26
4.2 Experiment 4 – Professional Documentation 26

1 Experiment 3 – Web & Connectivity

1.1 Learning Objectives

After studying this experiment, students should be able to:

- Understand basic web and Internet concepts.
- Use a web browser to access websites.
- Understand the difference between a web browser and a search engine.
- Perform advanced searches using search operators.
- Use the `filetype:` and `site:` search operators.
- Understand basic email etiquette.
- Write a professional email.
- Understand the purpose of Google Classroom/LMS.
- Enroll in a Google Classroom or LMS and customize the profile.

Important

The practical outcomes of this experiment include advanced search operators, email etiquette, and Google Classroom.

1.2 What is the Internet?

Definition

The **Internet** is a large network that connects computers and other digital devices around the world.

The Internet allows users to:

- Access websites
- Send and receive emails
- Share information
- Download files
- Upload files
- Communicate with other people

- Use online learning platforms

Example

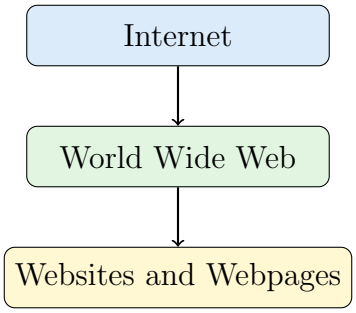
When a student opens a website to download study material, the Internet is used to transfer the information between the student's computer and the web server.

1.3 What is the World Wide Web?

Definition

The **World Wide Web (WWW)** is a collection of webpages and online resources that can be accessed through the Internet using a web browser.

The basic relationship can be represented as:



1.4 What is a Web Browser?

Definition

A **web browser** is a software application used to access and view websites and webpages.

Examples:

- Google Chrome
- Mozilla Firefox
- Microsoft Edge

Example

To open a college website, a browser such as Google Chrome can be used.

1.5 What is a Search Engine?

Definition

A **search engine** is an online service that helps users find information on the Web.

Example:
Google Search is a search engine.
A search engine can be used to find:

- Webpages
- Images
- Videos
- Documents
- PDF files
- Other online information

1.6 Web Browser vs Search Engine

Web Browser	Search Engine
Software application	Online search service
Used to access websites	Used to find information
Examples: Chrome, Firefox, Edge	Example: Google Search

Important

Chrome is a browser, not a search engine.
Google Search is a search engine, not a web browser.

1.7 What is Advanced Search?

Definition

Advanced search means using special search operators to make a search more specific and obtain more relevant results.

Two useful search operators are:

Operator	Purpose
`filetype:`	Search for a particular file type
`site:`	Restrict search to a particular website/domain

1.8 The `filetype:` Search Operator

Definition

The `filetype:` operator is used to search for a particular type of file.

Example:

```
filetype:pdf Computer Engineering
```

This tells the search engine to look for PDF files related to Computer Engineering.

Other examples:

```
filetype:pdf Python
```

```
filetype:ppt Artificial Intelligence
```

Important

For this experiment, the important example is `filetype:pdf`.

1.9 The `site:` Search Operator

Definition

The `site:` operator is used to restrict search results to a particular website or domain.

Example:

```
site:edu
```

This restricts the search to websites under the `.edu` domain.

Another example is:

```
site:edu Computer Engineering
```

1.10 Using Multiple Search Operators

Search operators can be combined to make a search more specific.

The prescribed search activity is:

```
filetype:pdf "Computer Engineering" site:edu
```

The three parts have the following meanings:

Search Term	Purpose
`filetype:pdf`	Searches for PDF files
`"Computer Engineering"`	Searches for the exact phrase
`site:edu`	Restricts results to the .edu domain

Therefore, the search asks for PDF files containing the phrase "Computer Engineering" from the `.edu` domain.

1.11 Use of Quotation Marks in Search

Quotation marks can be used to search for an exact phrase.

Without quotation marks:

```
Computer Engineering
```

With quotation marks:

```
"Computer Engineering"
```

The second search asks the search engine to look specifically for the phrase "Computer Engineering".

Example

Another example is:

```
"Artificial Intelligence"
```

This searches for the exact phrase "Artificial Intelligence".

1.12 Practical Activity – Advanced Search

Practical Activity

Open a web browser and search:

```
filetype:pdf "Computer Engineering" site:edu
```

Record the following:

1. First PDF result
2. Title of the PDF
3. URL
4. File size, if available

Important

The search result should be checked before using it. The result should actually be a PDF and should satisfy the search conditions.

1.13 What is Email?

Definition

Email, or electronic mail, is a method of sending and receiving messages electronically through the Internet.

Email can be used to:

- Communicate with teachers
- Submit information
- Contact organizations
- Send documents
- Receive notifications
- Communicate professionally

1.14 What is Email Etiquette?

Definition

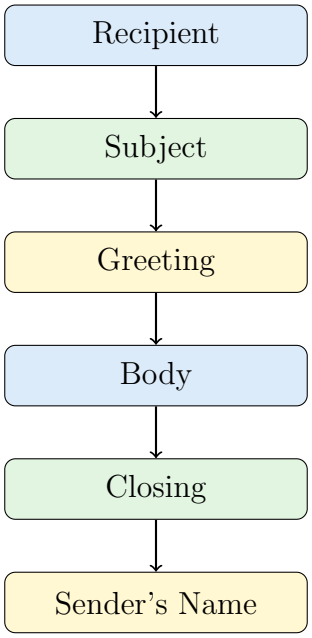
Email etiquette means following appropriate and professional rules when writing and sending emails.

A professional email should be:

- Clear
- Polite
- Concise
- Properly formatted
- Relevant to the purpose

1.15 Structure of a Professional Email

A professional email generally contains:



1.16 Recipient

The **recipient** is the person who will receive the email.

Example:

```
To: teacher@college.edu
```

The actual institutional email address should be used when sending an official email.

1.17 Subject

The **subject** briefly tells the receiver what the email is about.

Poor subject:

```
Hi
```

Better subject:

```
Request for Permission to Attend Technical Event
```

A good subject helps the receiver understand the purpose of the email quickly.

1.18 Greeting

A professional email should begin with an appropriate greeting.

Examples:

```
Dear Sir,
```

```
Dear Madam,
```

1.19 Body

The body contains the main message.

It should clearly state:

1. The reason for writing.
2. Necessary information.
3. The request or action required.

The message should be written politely and clearly.

1.20 Closing

A professional email can end with:

```
Regards,
```

or

```
Thank you,
```

followed by the sender's name.

1.21 Example of a Professional Email

Example

Subject: Invitation to Technical Event

Dear Sir/Madam,
We are pleased to invite you to attend the technical event organized by our college. The event will provide students with an opportunity to learn about current developments in technology.
We kindly request you to attend the programme.

Regards,
Event Coordinator

1.22 Common Email Mistakes

The following should be avoided in professional emails:

1. Unclear subject.
2. Text-message language.
3. Excessive capital letters.
4. Excessive emojis.
5. Sending without checking the recipient.

Example of text-message language:

```
pls send me the details
```

Instead:

```
Please send me the details.
```

1.23 Email Activity

Practical Activity

Scenario:
A student acts as the **Event Coordinator of the college**.
The student has to prepare and send a formal email inviting students to a technical event.
The email should contain:

- Appropriate recipient
- Suitable subject
- Professional greeting
- Purpose of the event
- Relevant information
- Appropriate closing
- Name/designation

1.24 What is an LMS?

Definition

LMS stands for **Learning Management System**. An LMS is a platform used to manage teaching and learning activities online.

An LMS can be used to:

- Share learning materials
- Give assignments
- Submit work
- Communicate with students
- Organize online classes

1.25 Google Classroom

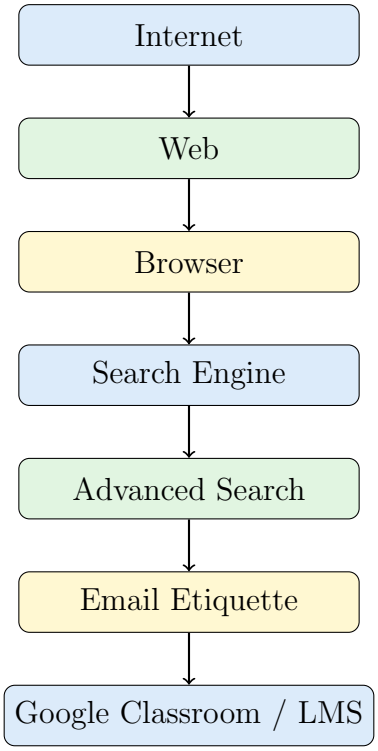
Definition

Google Classroom is an online learning platform that can be used by teachers and students to manage classroom activities digitally.

For this experiment, students are expected to:

1. Enroll in Google Classroom or another LMS.
2. Customize their profile.
3. Access the class and learning activities.

1.26 Summary – Web & Connectivity



Important

- A browser is used to access websites.
- A search engine is used to find information.
- `filetype:` searches for a particular file type.
- `site:` restricts search results to a particular domain.
- Quotation marks search for an exact phrase.
- Professional emails require appropriate formatting and language.
- An LMS supports online teaching and learning.

2 Experiment 4 – Professional Documentation

2.1 Learning Objectives

After studying this experiment, students should be able to:

- Understand professional documentation.
- Create a simple formal report.
- Organize information using headings.
- Apply document styles.
- Understand Heading 1 and Heading 2.
- Create a Table of Contents.
- Understand why styles are useful in long documents.

2.2 What is a Document?

Definition

A **document** is a collection of written or digital information prepared for a particular purpose.

Examples include:

- Letters
- Reports
- Assignments
- Project reports
- Lab records
- Notices

2.3 What is Professional Documentation?

Definition

Professional documentation means preparing information in a clear, organized and professional format.

A professional document should be:

- Clear
- Well organized
- Consistent
- Easy to read
- Properly formatted

2.4 Formal Report

Definition

A **formal report** is a structured document prepared to present information about a particular topic, activity or investigation.

A simple report may contain:

1. Introduction
2. Objectives
3. Methodology
4. Results
5. Conclusion

The exact structure can vary depending on the type of report.

2.5 What are Styles?

Definition

A **style** is a predefined collection of formatting properties that can be applied to text.

A style can control properties such as:

- Font
- Font size
- Bold/normal appearance
- Spacing
- Alignment

- Other formatting properties

Common styles include:

```
Title
Heading 1
Heading 2
Normal
```

2.6 Why are Styles Important?

Suppose a report contains 20 headings.

If each heading is formatted manually, it takes a lot of time and the formatting may become inconsistent.

Using styles allows the same formatting to be applied consistently.

Without Styles	**With Styles**
Manual formatting	Consistent formatting
Time-consuming	Faster to apply
May become inconsistent	Consistent throughout document

Important

Styles provide consistency and make document management easier.

2.7 Heading Styles

Heading styles are used to identify the structure of a document.

Example:

```
Heading 1
    Introduction

Heading 1
    Methodology

Heading 2
    Data Collection

Heading 2
```

```
    Data Analysis

Heading 1
    Conclusion
```

Here:

- **Heading 1** represents major sections.
- **Heading 2** represents subsections within a major section.

2.8 Manual Formatting vs Heading Style

Suppose we have:

```
Introduction
```

We can make it bold and increase its size manually.

However, simply changing its appearance does not necessarily tell the word processor that it is a heading.

Instead, applying:

```
Heading 1
```

identifies it as a major heading.

Important

Proper heading styles are important because features such as an automatic Table of Contents depend on the document structure.

2.9 What is a Table of Contents?

Definition

A **Table of Contents (TOC)** is a list of the main sections and subsections of a document, usually with their corresponding page numbers.

Example:

```
TABLE OF CONTENTS

1. Introduction ..... 1
2. Objectives ..... 2
3. Methodology ..... 3
4. Results ..... 6
5. Conclusion ..... 9
```

2.10 Why is a Table of Contents Useful?

A Table of Contents helps the reader:

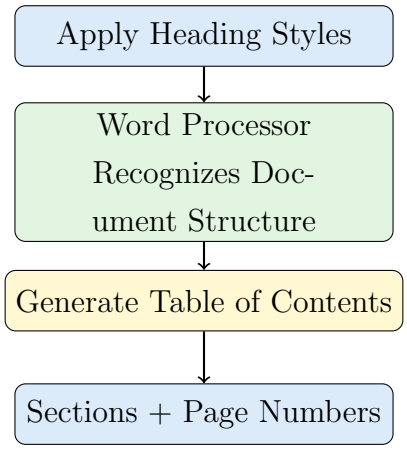
- Understand the structure of the document.
- Find a particular section quickly.
- Navigate through a long document.
- Know where different topics are located.

It is especially useful in:

- Project reports
- Technical reports
- Research reports
- Long assignments
- Manuals

2.11 Relationship Between Styles and Table of Contents

The relationship can be represented as:



Important

Proper use of heading styles makes it possible to generate an automatic Table of Contents.

2.12 Creating a Simple Formal Report

A simple report can be organized as follows:

```
MY FIRST TECHNICAL REPORT

1. Introduction

2. Objectives

3. Methodology

4. Results

5. Conclusion
```

Formatting

- Apply **Title** style to the report title.
- Apply **Heading 1** to the main sections.
- Apply appropriate lower-level heading styles to subsections.
- Insert a Table of Contents.

2.13 Practical Activity – Professional Documentation

Practical Activity

Task:

Create a document titled:

My First Technical Report

Include the following sections:

```
1. Introduction
2. Objectives
3. Hardware and Software
4. Internet Applications
5. Conclusion
```

Requirements:

1. Add a suitable title.
2. Apply proper heading styles.
3. Organize the sections properly.
4. Insert a Table of Contents.
5. Maintain consistent formatting.
6. Save the completed document.

2.14 Practical Procedure

Step 1 – Open the Word Processor

Open the word-processing application available in the laboratory.

Step 2 – Enter the Title

Type:

```
My First Technical Report
```

Apply the **Title** style.

Step 3 – Enter Main Headings

Type:

```
1. Introduction
```

```
2. Objectives
3. Hardware and Software
4. Internet Applications
5. Conclusion
```

Step 4 – Apply Heading 1

Apply **Heading 1** style to each main section.

Step 5 – Insert Table of Contents

Use the word processor's Table of Contents feature to generate the TOC.

Step 6 – Check the Document

Check:

- Headings
- Spelling
- Page numbers
- Table of Contents
- Overall formatting

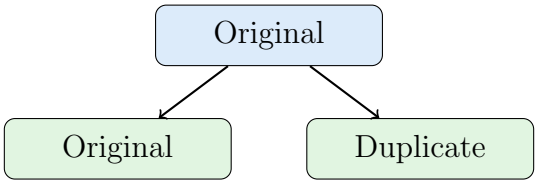
Step 7 – Save the Document

Save the completed report with a suitable filename.

2.15 Copy and Move

Definition

Copy creates another copy of a file while keeping the original.

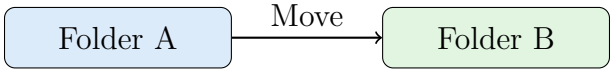


Therefore:

Copy = Duplicate

Definition

Move changes the location of the original file.



Therefore:

Move = Change Location

2.16 Common Mistakes in Professional Documentation

1. **Formatting headings manually**

   Instead of manually making headings bold and large, use the appropriate heading style.

2. **Inconsistent headings**

   For example:

   ```
   INTRODUCTION → 18 pt
   OBJECTIVES   → 14 pt
   RESULTS      → 20 pt
   ```

   Use consistent styles instead.

3. **Creating a TOC manually**

   A manually typed TOC may become incorrect when page numbers change.

   Use the word processor's automatic Table of Contents feature.

4. **Poor organization**

   Information should be arranged under meaningful headings and sections.

5. **Not checking the final document**

   Always check spelling, headings, page numbers and overall appearance before submitting.

2.17 Combined Practical Activity

The two experiments can be connected through one simple activity.

Part A – Web Search

Search:

```
filetype:pdf "Computer Engineering" site:edu
```

Select a relevant PDF.

Part B – Documentation

Create a short report about the information obtained from the selected resource.

Organize the report using:

`Title`

`1. Introduction`

`2. Main Information`

`3. Findings`

`4. Conclusion`

Apply appropriate heading styles and insert a Table of Contents.

Important

Web skills help us find information, while documentation skills help us organize and present information professionally.

3 Quick Revision Questions

3.1 Web & Connectivity

1. What is the Internet?
2. What is the World Wide Web?
3. What is a web browser?
4. Give two examples of web browsers.
5. What is a search engine?
6. Give one example of a search engine.
7. What is advanced search?
8. What is the purpose of `filetype:pdf`?
9. What is the purpose of `site:edu`?
10. Why are quotation marks used in a search?
11. What is email etiquette?
12. List the main parts of a professional email.
13. What is an LMS?
14. What is Google Classroom?

3.2 Professional Documentation

1. What is a document?
2. What is professional documentation?
3. What is a formal report?
4. What is a style?
5. What is Heading 1?
6. What is Heading 2?
7. Why are styles useful?
8. What is a Table of Contents?

9. Why is a Table of Contents useful?
10. What is the relationship between heading styles and a Table of Contents?

4 Quick Summary

4.1 Experiment 3 – Web & Connectivity

Concept	**Key Point**
Browser	Used to access websites
Search Engine	Used to find information
`filetype:`	Searches for a particular file type
`site:`	Restricts search results to a particular domain
Quotation Marks	Search for an exact phrase
Email Etiquette	Rules for professional email communication
LMS	Platform for online teaching and learning
Google Classroom	Online classroom management platform

4.2 Experiment 4 – Professional Documentation

Concept	**Key Point**
Formal Report	Structured presentation of information
Style	Predefined formatting applied to text
Heading 1	Used for major sections
Heading 2	Used for subsections
Table of Contents	Lists sections and page numbers
Professional Documentation	Clear, organized and consistent presentation

Important

Key Things to Remember

- Browser ≠ Search Engine
- `filetype:pdf` → Search for PDF files
- `site:edu` → Restrict search to the specified domain
- Professional email → Subject + Greeting + Body + Closing
- Style → Consistent document formatting
- Heading styles → Identify document structure
- Table of Contents → Provides organized sections and page numbers